Pooled testing for HIV viral load: simulation of reaction use and time to confirmation of failure, with parameters from the Brazilian public network

Willian Wistuba Melo da Cunha
Independent researcher

Abstract. Background: in HIV treatment programs with high viral suppression, most routine viral load tests come back with no detectable virus. In Brazil, about 89% of results from people on treatment for six months or more are below 50 copies/mL, and the public network contracts around 1.1 million tests per year. Testing samples in pools saves reactions when most are negative, but dilution and low-level viremia may delay the detection of failure at the 200 copies/mL cutoff. This note records the technique and estimates its potential, using Brazilian parameters as a reference case. Methods: simulation with the national viral load distribution (2022) and with the detection, quantification and reproducibility described in the instructions for use of the kits used in the network (cobas HIV-1 on the 5800 and 4800 systems). Retest rules were restricted to those the kit allows to be executed: retest any detected pool, including "detectado, inferior a 20 cp/mL" (rule D), or only those quantified at 20 cp/mL or more (rule Q20). In the cross-sectional stage (400,000 collections), strategies were compared by subrange, with discordances in both directions. In the longitudinal stage, a cohort of 200,000 suppressed adults was followed for 24 months, with collections every six months and confirmation in a new collection 28 days after the report. Variation comes from 40 replicates. Results: in the main scenario, pools of 4 with rule D used 37.9% fewer reactions per collection (empirical interval across replicates 37.6% to 38.3%), with no case detectable by individual testing failing to go to retest. During follow-up, they used 48.5% fewer reactions (48.4% to 48.7%), with a simulated difference close to zero in confirmation within 60 days (0.0 point; -0.4 to 0.2) and a median of 47 days until confirmation, against 42 for individual testing. Rule Q20 saved 4 to 5 more points, with small losses just above the cutoff. Quantitative deconvolution saved 57% of reactions, but detected only 38.5% of loads between 200 and 249, against 65.5% for individual testing. Pre-concentration of plasma brought no gain. Conclusion: in this simulation, the technique reduces reactions by 38% to 49% with no measurable loss in the detection of failure. If laboratory validation of the procedure, its admission into the execution rules and payment that reflects the number of reactions converge, the cost reduction compared with current practice would be a fraction of these percentages, proportional to the weight of the reaction in the price per test: for example, 11% to 29% of the annual value for weights of 30% to 60%.

Keywords: HIV viral load; pooled testing; simulation; virological failure; test costs; health informatics.

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https://mais-saude-ai.vercel.app/en/notes/testagem-grupo-carga-viral-hiv/

Translated from the Portuguese original.

In plain language

Why did we do this study? People who live with HIV and take treatment repeat a blood test a few times a year to see if the virus is under control. Almost always the result says there is no virus. We wanted to know if it is possible to use fewer tests without taking longer to find out when the treatment stops working.

What did we do and what did we find? We used a computer to try out an old idea: mix a little of the blood of four people and test the mix only once. If the mix shows no virus, no one in the group needs a new test. If it does, each person is tested alone. We used the numbers of the test used in Brazil. The idea would use between a third and a half fewer tests and would find the same problem cases, with about five more days of waiting. We also tried a bolder idea, and it did not help.

What does this mean? The idea works well on the computer, but it still needs to be tried in a lab. To turn into money saved, rules and the way the tests are paid for would also need to change. The note keeps the numbers on record for when this is possible. It says nothing about the health of any person.

I. INTRODUCTION

A. The problem and why it matters

The viral load test is the main way to know whether HIV treatment is working. The more effective the treatment, the larger the share of routine tests with no detectable virus. In Brazil, in 2022, 89% of people on treatment for six months or more had fewer than 50 copies/mL (Table II), and the public network works with about 1.1 million HIV viral load tests per year, at a reference price of R$ 67.31 per test in 2026 [15].

When most samples are negative, testing them in pools can save reactions. The technical question is whether this is possible without delaying the detection of those in failure, which, under the operational rule used in this note, means two consecutive results of 200 copies/mL or more, with at least 4 weeks between collections [12].

B. What is already known

Pooled testing was proposed in the 1940s [1] and has been used for decades in blood banks, including with plasma concentration by centrifugation to offset dilution [2].

For HIV viral load during treatment, studies from 2009 to 2017 evaluated pools of 5 and 10 × 10 matrices with quantitative deconvolution, with reductions from one third to more than half in the number of tests [3–6]. There are also prediction-guided methods [7]. In 2025, a reference laboratory in Cameroon reported routine use of pools of 3, with 0.55 assays per result [8].

These works were found through a general internet search and by reading abstracts, on Nov 8, 2026. There was no systematic literature search: PubMed, SciELO, LILACS, Europe PMC and other databases were not searched systematically, and the full texts were not read. For this reason, this note does not claim that the approach is new. As far as the search reached, many earlier studies prioritized thresholds higher than 200 copies/mL and used kits that predate the current ones.

C. The question

This note asks, with Brazilian public parameters:

- **Performance per collection:** how many reactions does each pooling strategy save, and how does it compare with individual testing in detecting loads between 200 and 249, 250 and 299, 300 and 499 and 500 and 999 copies/mL?
- **Performance during follow-up:** among suppressed adults, does the best strategy confirm virological failure within 60 days in the same proportion as individual testing, and with how much delay?
- **First principles:** does concentrating the pool's plasma, to avoid dilution, improve the result?
- **Potential:** if the conditions for adoption converged, how large would the cost reduction be compared with current practice?

D. What this note does and does not do

The note records the technique, simulates its performance and estimates its economic potential in a conditional way. The network's procurement documents are used only as a reference for the scale, price and payment model of current practice. The note does not test real samples, does not assess laboratories, contracts or suppliers, does not guide the follow-up of any person and does not replace bench validation.

II. DATA

Below the lowest concentration with data (50 cp/mL on the 5800 and 77 on the 4800), the standard deviation was assumed to be 0.25 log10 at 20 cp/mL. The robustness scenario used 0.40 below 50 cp/mL, 0.20 at 200 and 0.12 from 1,000 upward.

III. METHODS

A. Design overview

There are two stages, with the same assay parameters. The cross-sectional stage compares strategies in a single round of collections. The longitudinal stage follows people over time and measures the time until confirmation. A third part converts the reduction in reactions into economic potential, in a conditional way. All code uses fixed seeds. The note has no protocol registered before the analyses, and all results are exploratory.

B. Measurement model

The chance that the kit detects the virus in a reaction with concentration c was modeled as

$$P(\text{detect}) = 1 - e^{-3c/L} \tag{1}$$

where L is the 95% limit of detection: with $c = L$, the chance is $1 - e^{-3} \approx 95\%$. The model reproduces the detection rates in the instructions for use at 20, 10 and 5 cp/mL with a difference of up to 2 points (Table III) and overestimates detection at 2.5 cp/mL (43% against 35%), which only sends some pools to unnecessary retest. The measured value follows a lognormal distribution around the true value, with the standard deviation in Table III.

C. Reference

Individual testing. Each sample is tested once; the result is elevated when the measurement is 200 cp/mL or more.

D. Pooling strategies

The pool receives equal parts of k samples. The retest rules were restricted to those executable with the kit's output:

- **Rule D:** the pool goes to retest whenever the kit detects virus, including when the result is "detectado, inferior a 20 cp/mL", with no numerical value.
- **Rule Q20:** the pool goes to retest only when it is quantified at 20 cp/mL or more. In a pool of 4, this is equivalent to 80 cp/mL per sample.

Numerical triggers below 20 cp/mL in the pool were not used, because the kit does not report a value in that range. The retest is a new measurement of the stored sample, independent of the measurement in the individual arm. Pools of 3, 4, 5 and

Table I. Public sources used, with period and main limitation.

Source	What it provides	Period	Main limitation
HIV Clinical Monitoring Report 2022, Figure 76 [9]	national distribution of viral load by range	Jan to Sep 2022	includes everyone on treatment for 6 months or more, not only routine tests of suppressed people; values read from a chart
Instructions for use of cobas HIV-1 for cobas 5800/6800/8800 [10]	detection near the limit, limit of quantification, reproducibility, minimum volume, how results are reported	version consulted on Nov 8, 2026	precision measured on the 6800/8800 systems; no imprecision data below 50 cp/mL
Instructions for use of cobas HIV-1 for cobas 4800 [11]	the same items for the 4800	version consulted on Nov 8, 2026	no imprecision data below 77 cp/mL
Technical Notes 286/2023 and 74/2026 [12, 13]	network kits and result categories; operational definition of confirmed failure	2023 to 2026	summary of the instructions for use; genotyping context
Network procurement documents [14–17]	annual scale, reference price, payment model and execution rules of current practice	2021 to 2026	contracted quantities, not performed

Table II. Viral load distribution used in the simulations: people aged 2 years or older, on treatment for at least six months, Brazil, January to September 2022 [9].

Range (copies/mL)	Proportion
below 50	89%
50 to 199	4%
200 to 999	2%
1,000 to 9,999	2%
10,000 or more	3%

8 were evaluated with rule D, and of 4 with rule Q20. Two comparisons complete the set:

- **Quantitative deconvolution** with pools of 5 and a threshold of 250 cp/mL (configuration simulated from [3, 4]): if the pool indicates 250 or more per sample, the samples are tested one by one, and the value of each is subtracted from the estimated total until the remainder falls below the threshold. Only quantified values enter the calculations.

- **Pre-concentration:** 1 mL from each of 4 samples is concentrated into the volume of one reaction, with mean recovery of 0.7 and 0%, 2% or 5% of poor recoveries (between 0 and 0.2), with rule D.

Example

In a pool of 4 with one sample at 240 cp/mL and three with no virus, the pool has about 60 cp/mL. The kit detects and quantifies it, and the pool goes to retest under both rules. If imprecision makes the pool measure 15, the result comes out "detectado, inferior a 20": the pool goes to retest under rule D and does not under rule Q20.

E. Cross-sectional stage

400,000 collections were simulated with the distribution in Table II, drawn uniformly on a log scale within each range. For each strategy, reactions per sample and the share detected as 200 or more in each subrange were measured. Discordances with individual testing were counted in both directions, with independent measurements in both arms. The loss caused by the retest rule was separated out: cases that individual testing detects and whose pool did not go to retest.

F. Longitudinal stage

A cohort of 200,000 suppressed adults was followed for 24 months, with routine collections at months 6, 12, 18 and 24.

- Sustained rebounds arise with an annual risk of 6%. Two in seven stay between 200 and 999 cp/mL, and the rest above 1,000, in the proportion of Table II.

- Transient viremias between 50 and 199 cp/mL appear in 4% of collections, as in Table II.

- Each elevated collection triggers a new collection 28 days after the report, always tested individually. Two results of 200 or more confirm failure, and the person leaves the pools.

- The report takes 7 days, plus 5 days when the pool goes to retest.

The operational rule follows Technical Note 74/2026 ($\geq$ 200 cp/mL); the 2024 PCDT uses the wording "$>$ 200", and the difference does not change the simulation.

Primary outcome: proportion of episodes confirmed within 60 days after the first routine collection with a true load of 200 or more, counting in the denominator the episodes that the strategy missed. **Secondary outcomes:** confirmation by the

Table III. Assay parameters used in each scenario, from the instructions for use.

Parameter	cobas 5800 (main scenario)	cobas 4800 (sensitivity)
Processing volume	500 μL	400 μL
Minimum volume in the tube	650 μL	1,000 μL (secondary tube)
Limit of detection (95%)	13.2 cp/mL	14.2 cp/mL
Detection at 20, 10 and 5 cp/mL	98.4%, 90.5% and 66.1%	98.0%, 90.1% and 63.5%
Limit of quantification	20 cp/mL	20 cp/mL
Result below quantification	"HIV detetado, inferior a 20 cp/ml"	"HIV detectado, inferior a 20 cp/ml"
Standard deviation (log10) used	0.191 at 50; 0.116 at 200; 0.102 at 400; 0.092 at 1,000; 0.081 to 0.084 above	0.19 at 77; 0.09 at 308 and 770; 0.06 at 7,700; 0.05 above
Source of the standard deviation	reproducibility in 3 centers (6800)	within-laboratory precision

next routine collection (second opportunity, six months later) and median days until confirmation. Episodes whose first elevated collection was up to month 18 are included.

G. Economic potential

The reduction in reactions does not convert fully into a cost reduction, because the price per test also includes equipment, collection, transport, maintenance, technical support and training [16, 17]. The potential was estimated as

$$\text{annual reduction} = R \times w \times V \tag{2}$$

where R is the reduction in reactions, w is the weight of the reaction in the price per test and V is the annual reference value (1.1 million tests at R$ 67.31, about R$ 74 million [15]). As w is not public, illustrative values of 30%, 50% and 60% were used. For example, with $R = 38\%$ and $w = 30\%$, the reduction would be 11.4% of the annual value.

H. Variation

The intervals are empirical across 40 replicates with different seeds (2.5% to 97.5%). They describe the variation under the simulated model, not the uncertainty of the parameters, which was explored through the scenarios and the sensitivity analysis.

I. Sensitivity analyses

- cobas 4800 parameters instead of the 5800.
- Robustness scenario with greater imprecision (Table III).
- Annual rebound risk of 3%, 6% and 10%, crossed with transient viremia in 4% and 8% of collections.

J. Protocol deviations

Not applicable: the protocol has not yet been registered. The rules and configurations were revised during exploration, after earlier results had been examined, including through an external review that pointed out a trigger that could not be executed with the kit. In a confirmatory version, the main rule (D), the handling of non-quantifiable results and the 60-day outcome will be fixed before the run.

IV. RESULTS

How to read these results

- "Fewer reactions" compares with individual testing on the same collections.
- Detection is the share of samples in each subrange that received a result of 200 or more. Individual testing does not reach 100% near the cutoff because imprecision pushes part of the loads close to 200 below it.
- Differences of 1 to 3 points in a subrange are within the variation expected between two measurements of the same sample (Table V).

A. Performance per collection

Across the 40 replicates, the pool of 4 with rule D used 37.9% fewer reactions (empirical interval 37.6% to 38.3%), with a difference of 0.1 point (-0.8 to 1.2) in detection between 200 and 499 cp/mL. Larger pools saved less, because more pools went to retest: with 8 samples, 60% of them. Quantitative deconvolution saved more, but detected only 38.5% of loads between 200 and 249 and 83.2% between 300 and 499.

Discordances in both directions reflect the repeatability of the assay near the cutoff, which also appears between two individual measurements of the same sample. The loss caused by the retest rule was zero under rule D, in all subranges.

Table IV. Cross-sectional stage, main scenario (cobas 5800): fewer reactions and detection by subrange (cp/mL). In bold, the main strategy.

Strategy	Fewer reactions	200 to 249	250 to 299	300 to 499	500 to 999	1,000 or more
Individual	0%	65.5%	89.1%	99.1%	100%	100%
Pool of 3, rule D	37.0%	68.4%	88.0%	99.1%	100%	100%
Pool of 4, rule D	**37.6%**	**66.4%**	**89.6%**	**99.0%**	**100%**	**100%**
Pool of 4, rule Q20	41.6%	67.9%	85.7%	99.1%	100%	100%
Pool of 5, rule D	35.9%	64.0%	88.5%	99.2%	100%	100%
Pool of 8, rule D	27.5%	65.6%	88.3%	98.9%	100%	100%
Pool of 5, deconvolution (threshold 250)	56.7%	38.5%	58.4%	83.2%	95.0%	98.6%
Pre-concentration of 4, no failures	37.4%	67.9%	88.3%	99.3%	100%	100%
Pre-concentration of 4, 5% failures	37.6%	66.3%	88.0%	99.0%	99.9%	100%

Table V. Discordances between individual testing and the pool of 4, with independent measurements in both arms, main scenario. The last column is the loss caused by the retest rule.

Rule and subrange (cp/mL)	Samples	Only individual detects	Only pool detects	Individual detects and pool not retested
D, 200 to 249	1,106	247	220	0
D, 250 to 299	894	83	71	0
D, 300 to 499	2,567	16	13	0
D, 500 to 999	3,512	0	0	0
Q20, 200 to 249	1,106	249	219	2
Q20, 250 to 299	894	83	71	0
Q20, 300 to 499	2,567	16	13	0

B. Pre-concentration

Pre-concentration brought no gain in reactions or in detection. Dilution was not what limited the savings: with a limit of detection of 13 to 14 cp/mL, a pool of 4 takes a sample of 200 cp/mL to about 50 in the pool, which is detected almost always. The limit came from viremia between 50 and 199 cp/mL, which sends pools to retest without indicating failure.

C. 24-month follow-up

Across the 40 replicates, the pool of 4 with rule D used 48.5% fewer reactions (empirical interval 48.4% to 48.7%), with a difference of 0.0 point (-0.4 to 0.2) in confirmation within 60 days. The median of 47 days corresponds to the direct path: report in 7 days plus 5 for retest, new collection 28 days later and confirmation report in 7 days. The longitudinal savings are larger than the cross-sectional ones because the cohort only includes suppressed people, and those with confirmed failure leave the pools.

D. Sensitivity

In the robustness scenario, rule Q20 fell to 94.6% confirmation within 60 days, against 95.0% for individual testing.

In all combinations, the difference in confirmation within 60 days was 0.2 point or less. The reduction in reactions ranged from 32% to 54%, mainly depending on the frequency of transient viremia.

E. Plasma volume

In the 500 μL workflow of the cobas 5800, the tube needs at least 650 μL [10]. A pool of 4 would require about 163 μL from each sample, and a possible individual retest another 650 μL. Each collection would need to yield at least about 0.8 mL of plasma, plus a reserve for repetition. On the cobas 4800, with 1,000 μL in the secondary tube, it would be about 1.25 mL [11].

F. Conditional economic potential

These values would only materialize if four conditions converged, none of them present today:

- **Laboratory validation** of the procedure on the equipment in use, with real samples.
- **Admission of the procedure into the execution rules.** The current kits do not provide for sample pooling in their instructions for use [10, 11], and the most recent procurement requires an automated methodology with operator interventions restricted to those provided for by the manufacturer [17]. The procedure would need to enter the instructions for use or be admitted by the contracting party.
- **Payment that reflects the number of reactions.** In current practice, payment is made once per sample with a released result, regardless of the number of repetitions [16, 17]. In this model, fewer reactions do not reduce the amount paid, and the per-result payment model

Table VI. Longitudinal stage, main scenario, 17,294 episodes: fewer reactions, confirmation within 60 days and by the next routine collection, and median days until confirmation. In bold, the main strategy.

Strategy	Fewer reactions	Confirmed within 60 days	By the next collection	Median until confirmation
Individual	0%	97.0%	98.5%	42 days
Pool of 4, rule D	**48.5%**	**97.1%**	**98.6%**	**47 days**
Pool of 4, rule Q20	52.9%	97.0%	98.7%	47 days
Pool of 3, rule D	45.8%	97.0%	98.5%	47 days

Table VII. Main results in the three assay scenarios, pool of 4 with rule D: fewer reactions per collection, samples between 200 and 999 cp/mL without retest, fewer reactions during follow-up and confirmation within 60 days (individual and pool).

Scenario	Per collection	Without retest	During follow-up	Within 60 days
cobas 5800 (main)	37.6%	0	48.5%	97.0% and 97.1%
cobas 4800	37.6%	0	48.6%	97.1% and 97.2%
Robustness (greater imprecision)	37.6%	0	48.2%	95.0% and 95.1%

creates no economic incentive for those who perform the tests to adopt the technique. The reduction would only reach the contracting party through a specific price for samples tested in pools or through price competition in a procurement that admits the procedure.

- **Eligible tests and logistics:** share of routine tests from people already suppressed, sufficient plasma volume and adapted laboratory workflows.

Long-term contracts reduce the opportunities to revisit these conditions. Until they converge, the possible gain is operational: equipment and bench time freed up in the laboratories.

V. DISCUSSION

A. Main findings

In this simulation, pools of 4 with retest of any detected pool saved 38% of reactions per collection and 49% during follow-up of suppressed people, with no case that individual testing would detect failing to go to retest, and with a difference close to zero in confirmation of failure within 60 days. The cost was a median of 5 more days until confirmation. The result was repeated with both network kits and in the robustness scenario. Converted into cost, the reduction would be smaller than the reduction in reactions and would depend on conditions external to the technique.

Two results contradicted the study's own hypotheses. Preconcentration, designed to eliminate dilution, brought no gain, because dilution is not the bottleneck with the current kits. And rules with a higher trigger, which save a little more, start leaving cases just above the cutoff without retest when imprecision is greater.

B. Promising aspects and limits, side by side

This note does not use resampling of observed data or a registered success criterion; therefore, it does not calculate the chance of exceeding a reference. The promising aspects and the limits are these:

- **Promising:** no loss from lack of retest under rule D, in all subranges and in the three scenarios.
- **Promising:** difference of 0.2 point or less in confirmation within 60 days in all combinations of risk and transient viremia.
- **Promising:** a rule executable with the output the kit already delivers to the reporting systems.
- **Limit:** the intervals describe only the simulation's variation, not the uncertainty of the parameters.
- **Limit:** the reduction in reactions ranges from 32% to 54% depending on transient viremia and rebound risk, which have not been measured in routine tests of suppressed people in Brazil.
- **Limit:** the weight of the reaction in the price per test is not public, and the conversion into cost is illustrative.
- **Limit:** the technique is off-label use of the kit.
- **Limit:** the main configuration was chosen after comparing alternatives and after reviews, which requires confirmation with a registered protocol.

C. Comparison with the literature

Earlier studies reported from one third to more than half fewer reactions [3–6], and the implementation in Cameroon reached 0.55 assays per result with pools of 3 [8]. The 38% to 49% in this simulation fall within that range. Many earlier studies prioritized thresholds higher than the Brazilian one,

Table VIII. Longitudinal sensitivity, main scenario, pool of 4 with rule D: annual rebound risk, share of collections with transient viremia, fewer reactions and confirmation within 60 days (individual and pool).

Rebound	Transient viremia	Fewer reactions	Within 60 days
3%	4%	54%	96.9% and 96.9%
3%	8%	42%	96.9% and 97.1%
6%	4%	49%	97.0% and 97.1%
6%	8%	37%	96.9% and 96.9%
10%	4%	42%	96.7% and 96.9%
10%	8%	32%	96.9% and 96.8%

Table IX. Estimated annual cost reduction (R $\times$ w $\times$ V), with V of about R$ 74 million, for the reduction in reactions per collection (38%) and during follow-up of suppressed people (49%), depending on the weight of the reaction in the price (w).

Reaction weight (w)	With 38% fewer reactions	With 49% fewer reactions
30%	11.4% (R$ 8.4 million)	14.7% (R$ 10.9 million)
50%	19.0% (R$ 14.1 million)	24.5% (R$ 18.1 million)
60%	22.8% (R$ 16.9 million)	29.4% (R$ 21.8 million)

although the evaluation by Smith and colleagues included the 250 cp/mL cutoff [3]. With the 200 cutoff and the current kits, the simple rule of retest on any detection performed better than quantitative deconvolution, which missed a large part of the loads just above the cutoff. Pre-concentration has an operational precedent in blood banks [2] and in research with volumes of up to 30 mL [18], but, in this simulation, it did not prove necessary.

D. Strengths

- Assay parameters taken from the instructions for use of the two network kits, with the detection model checked against them.
- Retest rules restricted to those the kit allows to be executed.
- Detection broken down by subrange and discordances in both directions.
- A longitudinal outcome that counts missed episodes in the denominator and separates bench delay from delay due to loss.
- Explicit separation between reduction in reactions and reduction in cost, with the conditions for the latter.
- A first-principles hypothesis tested and reported even with a negative result.
- Code and numbers reproducible with fixed seeds.

E. Weaknesses of the method

- **Denominator.** The distribution in Table II includes everyone on treatment for six months or more. The technique would apply to routine tests of adults already suppressed, whose distribution was not found. With less low-level viremia, the reduction in reactions tends to be larger; with more, smaller.
- **Shape within ranges.** The uniform draw on a log scale within each range is an assumption; detection between 200 and 249 depends on it.
- **Imprecision at very low loads.** There are no data below 50 cp/mL on the 5800 or below 77 on the 4800; the value used is assumed. The robustness scenario suggests a small effect on rule D.
- **Dynamics.** Rebounds are sustained and at a constant level. Transient rebounds above 200, intermittent adherence failures and regimen changes are not included.
- **Laboratory.** Cross-contamination between samples, pipetting error, handling cost, bench time beyond the 5 days and adaptation of the reporting systems are not included. All of them tend to reduce the real gain.
- **Cost.** The cost structure of the price per test is not public; the reaction weight was assumed.
- **Choice after looking.** The main configuration emerged from exploration and reviews; confirmation requires a registered protocol.

F. What the result allows us to conclude

It allows us to conclude that, with the public parameters available, a simple strategy of pools of 4 with retest of any detected pool is compatible with detecting failure at the 200 cp/mL cutoff and with the confirmation time frame, saving 38% to 49% of reactions. It allows us to rule out, in this configuration, pre-concentration as a necessary improvement. It allows us to estimate, conditionally, a cost reduction of 11% to 29% of the annual reference value. It does not allow us to claim performance in a real laboratory, nor to claim that this

reduction will occur: it depends on validation, on admission of the procedure and on payment linked to the number of reactions, conditions absent from current practice.

G. Next steps

Each path below only becomes work with a new protocol:

- obtain the viral load distribution in routine tests of people who were already suppressed and the proportion of eligible tests;
- obtain imprecision data below 50 cp/mL;
- look for public estimates of the cost structure of viral load tests, to replace the illustrative reaction weight;
- include transient rebounds, plasma volume per sample and bench failures;
- carry out the systematic literature search, including validation studies of pooling on automated platforms;
- register the protocol before a confirmatory run.

VI. CONCLUSION

In this simulation with Brazilian public parameters, pools of 4 samples, with retest of any detected pool, used 37.9% fewer reactions per collection and 48.5% fewer during follow-up of suppressed adults. No case that individual testing would detect failed to go to retest, the difference in confirmation of failure within 60 days was close to zero (0.0 point; empirical interval -0.4 to 0.2) and the median until confirmation was 47 days, against 42. Retesting only quantified pools saved a little more, with small losses just above the cutoff. Quantitative deconvolution saved even more, but missed a large part of the loads between 200 and 499. Pre-concentration of plasma brought no gain. If laboratory validation, admission of the procedure into the execution rules and payment linked to the number of reactions converged, the cost reduction compared with current practice would be 11% to 29% of the annual reference value, depending on the weight of the reaction in the price. The note records the technique and these calculations for future studies.

VII. LIMITATIONS

- The research steps (literature search, reading of documents, code, simulation, analysis and writing) were carried out with artificial intelligence and did not undergo independent human validation. The results may contain errors or gaps because of this.
- The design weaknesses listed in the Discussion also count as limitations.
- The note has no protocol registered before the analyses, and the configurations were chosen during exploration.
- The instructions for use validate the test for EDTA plasma (and, in the case of the 4800, also for the plasma separation card), without providing for sample pooling. Pooled testing would be off-label use.
- The distribution in Table II was read from a chart, by enlarged visual inspection.
- The annual reference value uses the contracted quantity, not the quantity performed.

VIII. GLOSSARY

- **Cohort:** group of people followed over time; here, simulated.
- **Confirmed virological failure:** two consecutive results of 200 cp/mL or more, with at least 4 weeks between collections.
- **Discordance:** a sample that one strategy classifies as 200 or more and the other does not.
- **Empirical interval across replicates:** range of results obtained by repeating the simulation with different random draws.
- **Imprecision:** variation between repeated measurements of the same sample; here, standard deviation in log10.
- **Limit of detection:** lowest concentration detected in 95% of reactions.
- **Limit of quantification:** lowest concentration for which the kit reports a numerical value; below it, it reports only “detectado”.
- **Lognormal:** distribution in which the logarithm of the value follows the normal curve; used for measurement errors proportional to the value.
- **Off-label use:** use of a diagnostic kit in a way not provided for in the instructions approved by the manufacturer.
- **Pooled testing:** testing a mixture of several samples and retesting individually only when the mixture shows a signal.
- **Pre-concentration:** gathering more plasma from each sample and concentrating the virus before the reaction, to offset dilution.
- **Quantitative deconvolution:** testing the samples of a positive pool one by one, subtracting the value of each from the estimated total, until the remainder falls below a threshold.

- **Reaction:** each time the instrument processes a tube, whether of a single sample or of a pool.
- **Reaction weight in the price:** share of the price per test that corresponds to the cost of the reaction (reagents and run), excluding equipment, collection, transport and services.
- **Rule D:** retest individually the samples of every pool in which the kit detects virus.
- **Rule Q20:** retest only the samples of pools quantified at 20 cp/mL or more.
- **Seed:** number that fixes the sequence of random draws in a simulation, so that it can be repeated with the same result.
- **Transient viremia:** detectable, low viral load in a single collection, without treatment failure.
- **Viral load:** amount of HIV RNA per milliliter of plasma.

IX. REPORTING GUIDELINES

The note follows, insofar as it applies to simulation studies, the STRESS guideline [19] and, for the cost estimate, CHEERS 2022 [20]. The main items are in the sections: objectives and questions (I), sources and parameters (II), model logic, inputs, outcomes and variation (III), results with intervals, sensitivity analyses and economic potential (IV), limitations (V and VII) and reproducibility (Data and code availability).

DISCLAIMER

This is a working note in health informatics, produced with extensive use of artificial intelligence, and not a clinical study. The research steps carried out with AI (data collection, literature screening, analysis and code) have not undergone independent human validation, and the results may contain errors or gaps as a consequence; the author reviews the final text before publication. It has not been reviewed by a licensed health professional. It does not constitute advice, diagnosis, prognosis or treatment recommendation for any person. Hypotheses are hypotheses, not confirmed findings. The source data have limitations described in the text. Any use in health care requires specific review by licensed professionals, according to the intended use. Valid for the date and version indicated.

ARTIFICIAL INTELLIGENCE USE STATEMENT

Tools: Claude Opus 5.5 (Anthropic), in sessions on Nov 8 and 9, 2026. Steps with AI: formulation of the hypotheses, literature search, queries to the National Public Procurement Portal, reading of the public documents and of the kits' instructions for use, writing of all the code, running of the simulations, writing and translation into English and Spanish. Checking: automatic only: the numbers in the text were copied from the scripts' outputs, and the published scripts were run again to check that they reproduce those outputs; the detection model was compared with the tables in the instructions for use; excerpts from documents were checked by reading the PDFs. Reviews by another AI tool pointed out problems that were corrected before publication. There was no independent human validation of the steps carried out by AI, and the results may contain errors or gaps because of this; the author reviews the final text. Tool limitations: it may make mistakes in code, in reading figures and in reading literature without the error showing up in the automatic checks. Record: code and outputs versioned in the program's public repository (github.com/willianwistuba/mais-saude-ai), in codigo_src/msa02_*.py and dados/msa02/; there was no protocol registered before the analyses. Conflicts and funding: none.

DATA AND CODE AVAILABILITY

All data used are public, and the scripts that produce every number are available at https://mais-saude-ai.vercel.app/en/code/, with instructions to reproduce the results.

ETHICS

This note uses only public, aggregated data that do not identify individuals, which waives ethics committee review under Brazilian National Health Council Resolution 510/2016 and CONEP Circular Letter 17/2022. Counts below five are suppressed at municipal level, and no re-identification is attempted.

COMPETING INTERESTS AND FUNDING

The author declares no competing interests. This work received no funding, and the project accepts no resources from manufacturers of technologies it assesses or comments on. This text has not been peer reviewed, reviewed by a health professional or subjected to independent human validation of the research steps carried out with AI (the author reviews only the final text), and should be read as a working note. Corrections create a new version, and the previous one remains accessible.

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